

The Imitation Game and Modern Artificial Intelligence

The Imitation Game

How Modern AI is beating The Imitation Game by doing exactly that: Imitating

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CCS CONCEPTS • **Computing methodologies** • Artificial intelligence

1 INTRODUCTION

Alan Turing is one of the most influential computer scientists of all time. Turing's contributions to computer science include his work on *On Computable Numbers, with an Application to the Entscheidungsproblem* [1], which introduced the abstract Turing machine and established foundational concepts in computability. He later contributed to the development of the *Bombe* [2], which played an important role in Allied efforts to decipher Enigma-encrypted communications. With the 21st century's rise of large language models (LLMs) and artificial intelligence (AI), one of his most relevant papers is *Computing Machinery and Intelligence* [3], in which he examines whether machines are capable of thought. He does this by reframing the question with *The Imitation Game*, more commonly known today as *The Turing Test*. While this is a sound way to test whether a machine can *imitate* thought, it is not a sufficient test to determine whether AI is *capable* of thought.

1.1 Applicability to Artificial Intelligence

Artificial intelligence today is advancing rapidly, and therefore it is important to have some form of test to benchmark different models and to evaluate and compare their capabilities. While the *Turing Test* has been used to evaluate AI models in 2026 [4], it is insufficient on its own, and it does not adequately address the question that Turing asks: "Can Machines Think?"

Almost any model today can pass a basic Turing Test with relative ease [4]. The models are trained on how we speak and write. However, certain linguistic patterns can still distinguish AI-generated writing from human writing. These patterns may include frequently repeated phrases, writing styles, or punctuation conventions that are more characteristic of AI-generated text. Take this short exchange I had with OpenAI's GPT-5.6 Luna model:

Person A: Hey, how's it going?

Person B: Pretty good! Just winding down a bit. How about you? How's your night going?

Person A: Alright. Just working on a paper. You?

Person B: Same kind of thing, really—just hanging out and chatting. What’s the paper about?

At first glance, you would not be able to tell the difference until you look closer. One notable characteristic is the use of em-dashes. Because of patterns learned from their training data, models may utilize em-dashes more frequently than is typical in human writing. These characteristics highlight a fundamental issue when evaluating AI with the Turing Test: they generate responses through next-token prediction based on the preceding context. In other words, they *imitate* what a conversation would look like to an outside observer.

This illustrates a fundamental issue with the Turing Test: it does not test what Turing initially designed it to test. The AI above generates its response through next-token prediction; however, this on its own does not establish whether the AI is capable of thought. Another good example of this is with AI image generation models. The images produced by AI are based on patterns learned from training data, rather than knowledge of the physical scene, creating an *imitation* of what the scene looks like. They can contain inconsistencies that arise from the model generating an approximation based on learned visual patterns rather than directly representing the physical world. For example, a billboard in an AI-generated image might have words on it that appear plausible, but upon further inspection the words are unintelligible or the company being advertised does not exist. The following images illustrate this phenomenon (*Figure 1*).



Figure 1: An image taken from the third floor of UTSA’s San Pedro 1 Building (Left), and an image generated by Google’s Nano Banana of the same angle (Right)

In the AI-generated image, the model generates an approximation of the surrounding area based on patterns present in the input photo. In reality, there is no shorter building in the foreground, and the right side of the second building from the bottom does not look like that. The brick building in the foreground is also an odd shape, ending abruptly before the road.

Another example of how an AI imitates and does not think is the following scenario: Imagine you are describing a dog to someone who has never seen a dog. You then have that person draw the dog. What they draw will be close to the real thing, but some parts the person will have to fill in the blanks with a guess as to what goes there. An AI works the same way: it can be prompted to generate an image of a dog, but certain details will always feel off because it does not know how those details are in real life. It just imitates what a dog should look like. In both language and image generation, the AI can produce an output that appears convincing to an observer while containing evidence that it was generated from learned patterns rather than the kind of understanding the observer might assume.

1.2 A Misguided Goal

While the Turing Test can be used to benchmark a model’s ability to imitate a human, it is less useful as a goal for AI development. A model that can convincingly imitate a human is not necessarily useful, nor does it demonstrate that it can advance human knowledge. Rather than making an AI indistinguishable from a human, development should focus on how AI can extend those abilities and contribute positively to society.

One potential direction for achieving this goal is the development of artificial general intelligence (AGI), which aims to create an AI that is capable of performing a large range of tasks correctly and accurately. AGI can potentially contribute to discoveries in science, technology, and mathematics, helping humanity make progress beyond our current limitations. However, increased AI capabilities can also introduce significant risks, especially if they pursue objectives that conflict with human interests. AGI models that are not properly aligned with human interests could cause harm without intending to do so. The pursuit of AGI should not focus solely on increasing intelligence or accelerating development; it should also prioritize safeguards that ensure these systems remain beneficial to humanity. Rather than treating AGI as an achievement on its own, the goal should be to develop AI that advances human knowledge while protecting humanity and preserving human control.

1.3 LLMs are Complex Digital Computers

Functionally, LLMs operate as complex digital computers, taking an input, executing on it based on a set of instructions, and returning an output. Turing argues that digital computers can be constructed according to the principles he describes and can “mimic the actions of a human computer very closely” [3]. This distinction is particularly interesting when comparing mathematical equations with human conversation. A mathematical equation generally has an objectively verifiable result given a defined input and set of conditions, while human conversations can produce many reasonable responses to the same input. Different people may respond differently based on knowledge, experience, or personality. LLMs can similarly produce different responses to the same prompt, in part because their generation process allows for variation in the output. This variation can make their responses appear more natural, but it does not necessarily indicate that the model possesses the same thoughts, experiences, or reasoning processes as a human.

Turing further addresses this distinction when he asks whether a particular digital computer could be modified through sufficient storage, increased speed, and an appropriate program to “play satisfactorily the part of A in the imitation game” [3]. Modern AI systems demonstrate that this is possible to a degree that Turing could not have demonstrated in 1950. An LLM can be given an input, process that input using its parameters, and produce a response that can be convincing enough to resemble human conversation. In this sense, modern AI fits well within Turing’s proposed framework: a digital computer can be programmed to produce behavior that is difficult for an observer to distinguish from the behavior of a human. However, this ability still does not answer the underlying question of whether the computer is actually thinking. It demonstrates that a digital computer can produce outputs that imitate human behavior.

1.4 Objections

I would like to focus on a specific objection presented by Turing, “Lady Lovelace’s Objection”. In this, Lovelace states in regards to the *Analytical Engine*, “The Analytical Engine has no pretensions to *originate* anything. It can do *whatever we know how to order it to perform*” [3]. This is interesting in the context of modern AI, as current agentic systems are capable of performing complex tasks with less direct human input. For example, some frontier models, such as Claude Fable or GPT Astra, can create and manage sub-agents, determining what tasks those agents should perform

based on a broader objective without any human interacting with the sub-agent. This raises the question: if an AI is given a general objective and independently comes up with the steps necessary to accomplish it, is it originating something new?

Consider a system instructed to develop the next generation of AI capable of operating independently of human prompts. If that system designs a new model, develops new methods, and produces ideas without humans specifying the individual steps, it becomes difficult to argue that every resulting idea from the new model was directly “ordered” by a human. While there is still human influence in the initial instruction, individual outputs were not anticipated or programmed by the creators. This distinction will become more and more important as AI systems become increasingly autonomous. If an AI can independently learn, create new ideas, and develop new technology, it may challenge Lovelace’s claim that a machine can only do what humans know how to order it to do. This is also closely related to the idea of AGI: rather than simply being an AI capable of performing a wide range of tasks, AGI could eventually involve systems capable of independently learning, reasoning, and generating solutions without requiring humans to specify each step.

1.5 The Turing Award

Turing’s paper is an unusual academic work for its time. The concept of whether computers can think is one that was so unconventional that even he acknowledged that he does not have many convincing arguments in support, stating “The reader will have anticipated that I have no very convincing arguments of a positive nature to support my views. If I had I should not have taken such pains to point out the fallacies in contrary views.”[3] He also states, “As I have explained, the problem is mainly one of programming. Advances in engineering will have to be made too, but it seems unlikely that these will not be adequate for the requirements.” [3] Today, many of those advances in computer engineering Turing anticipated have become a reality.

Alan Turing has left a legacy of technological and mathematical advances, and he rightfully has an award named after him. While I don’t think that the Imitation Game is his best work, given how strong his other work and ideas were, his argument can be incomplete about one particular idea and he can still be deserving of the award.

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